

Teen Mental Health and Social Media Usage Analysis Report

Executive Summary

This study explores the relationship between social media usage, academic performance, sleep habits, physical activity, and stress levels among teenagers. The analysis identifies several behavioral patterns and correlations that provide valuable insights into how digital habits influence students' well-being and academic outcomes.

Objectives

- Analyze social media usage patterns among teenagers.
 - Understand the impact of social media on academic performance.
 - Examine the relationship between screen time, sleep duration, and stress levels.
 - Compare behavioral differences between male and female students.
 - Identify factors contributing to better academic and mental health outcomes.
-

Key Findings

1. Physical Activity by Gender

Male students exhibit higher levels of physical activity compared to female students.

2. Most Popular Social Media Platform

Instagram emerged as the most frequently used social media platform among teenagers.

3. Academic Performance by Age

Students aged 17 and 18 demonstrated the highest academic performance compared to other age groups. However, 17-year-old students also spent more time on social media than other age groups.

4. Social Media Usage and Academic Performance

Students who spend less time on social media tend to achieve higher academic performance, indicating a negative relationship between excessive social media usage and academic success.

5. Screen Time Before Sleep

Higher screen exposure before bedtime negatively impacts sleep duration, reducing the number of hours students sleep.

6. Stress Levels by Age

Teenagers aged 14 reported the highest stress levels among all age groups analyzed.

7. Stress Levels by Gender

Male students showed higher average stress levels compared to female students.

8. Social Media Usage by Gender

Male students spend more time on social media platforms than female students.

9. Social Media Interactions and Stress

The number of social media interactions (likes, comments, messages, etc.) showed little to no significant correlation with stress levels.

10. Sleep Duration by Gender

Female students generally sleep longer than male students.

11. Sleep and Academic Performance

Sleep duration is positively correlated with academic performance. Students who get adequate sleep tend to perform better academically.

12. Social Media Usage and Stress

Higher social media usage is associated with increased stress levels, suggesting that excessive engagement may negatively impact mental well-being.

Correlation Analysis

Variables	Relationship
Social Media Usage vs Academic Performance	Negative Correlation
Screen Time Before Sleep vs Sleep Duration	Negative Correlation
Sleep Duration vs Academic Performance	Positive Correlation

Variables	Relationship
Social Media Usage vs Stress Level	Positive Correlation
Social Media Interactions vs Stress Level	No Significant Correlation

Conclusion

The analysis reveals that excessive social media usage can adversely affect both academic performance and mental well-being. Students who spend less time on social media and maintain healthy sleep habits generally achieve better academic outcomes. Sleep duration plays a critical role in academic success, while increased social media usage contributes to higher stress levels. Gender and age differences also influence physical activity, stress, and sleep patterns among teenagers.

Recommendations

1. Encourage students to limit daily social media usage.
 2. Promote healthy screen-time habits before bedtime.
 3. Educate teenagers about the importance of adequate sleep.
 4. Implement stress-management programs, especially for younger teenagers.
 5. Encourage physical activities and outdoor engagement.
 6. Create awareness about maintaining a healthy balance between online activities and academic responsibilities.
 7. Parents and educators should monitor excessive social media usage and support healthy digital habits.
-

Final Insight

The findings suggest that maintaining a balanced lifestyle—including moderate social media use, sufficient sleep, and regular physical activity—can significantly improve both academic performance and mental health among teenagers.